Biochemistry Cume

December 2, 2011

1. *Escherichia coli* cells are growing in a medium containing lactose but no glucose. Explain how each of the following changes or conditions would affect the expression of the *lac* operon. **75 (5 x 15) points**

a. Addition of high concentration of glucose.

b. A mutation that prevents dissociation of the *lac* repressor from the operator.

c. A mutation that completely inactivates β-galactosidase.

d. A mutation that completely inactivates galactose permease.

e. A mutation that prevents binding of cyclic AMP Receptor Protein (CRP) to its binding site near the lac promoter.

2. The dissociation constant for a particular repressor-operator complex is very low, about 10^{-13} M. An *E.coli* cell (volume = 2×10^{-12} mL) contains 10 copies of the repressor. Calculate the cellular concentration of the repressor protein. (**25 points**)